

UESTCHN3007 Dynamics & Control
Lecture 2

Basics of Control System Analysis

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Outline

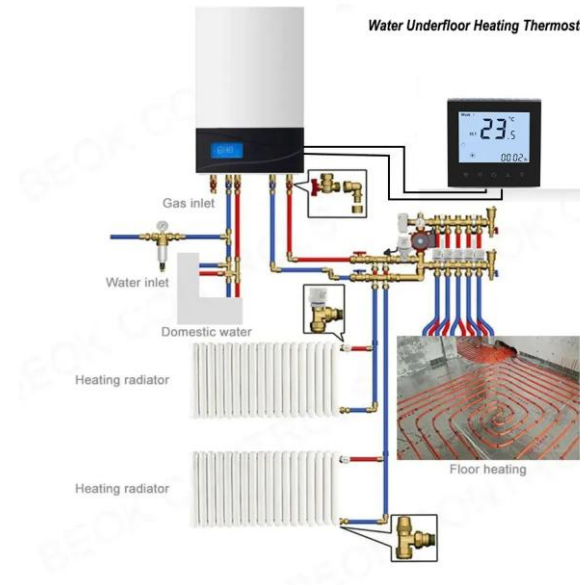
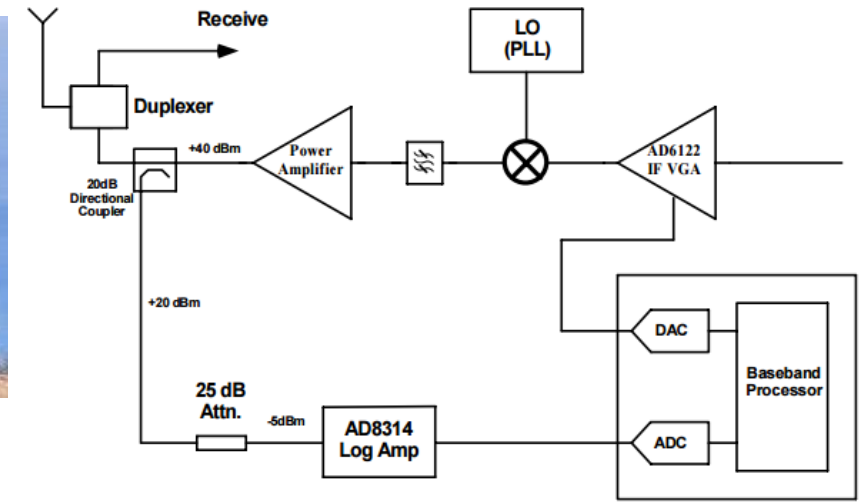
- Introduction to Control Systems
- Block Diagram Representation

Introduction to Control System

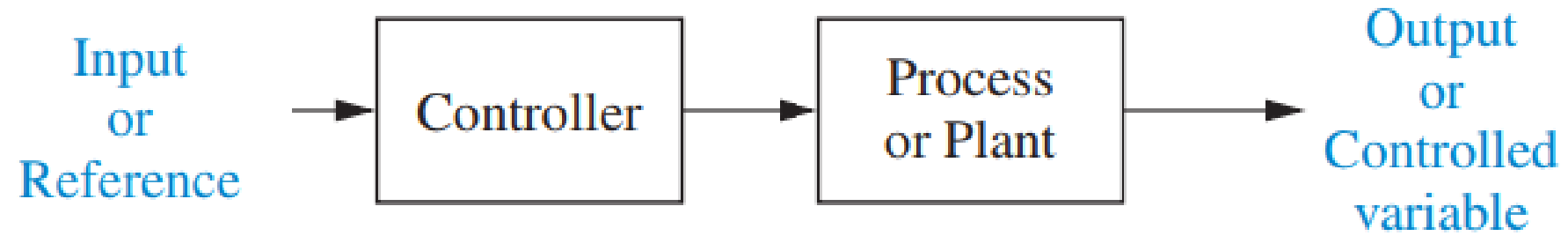


Advantages of Control Systems

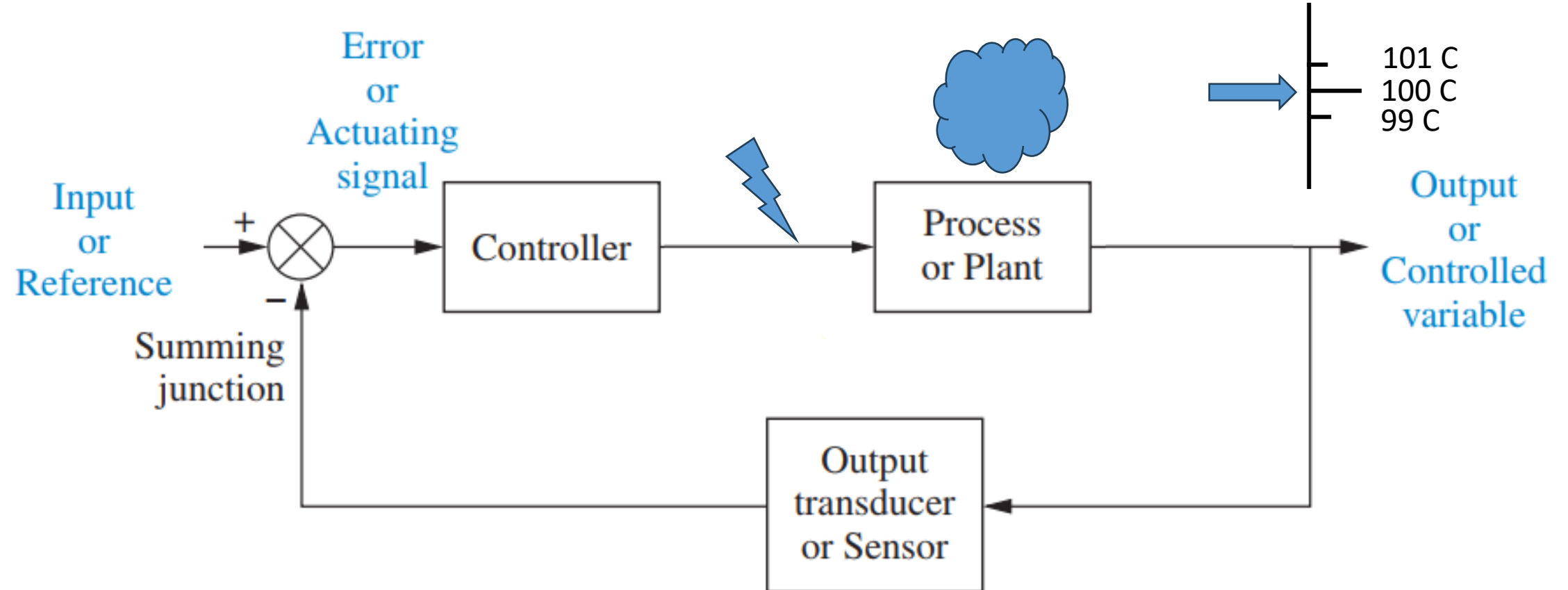
- Power amplification
- Remote control
- Convenience of input form
- Compensation for disturbances



Open-loop Control

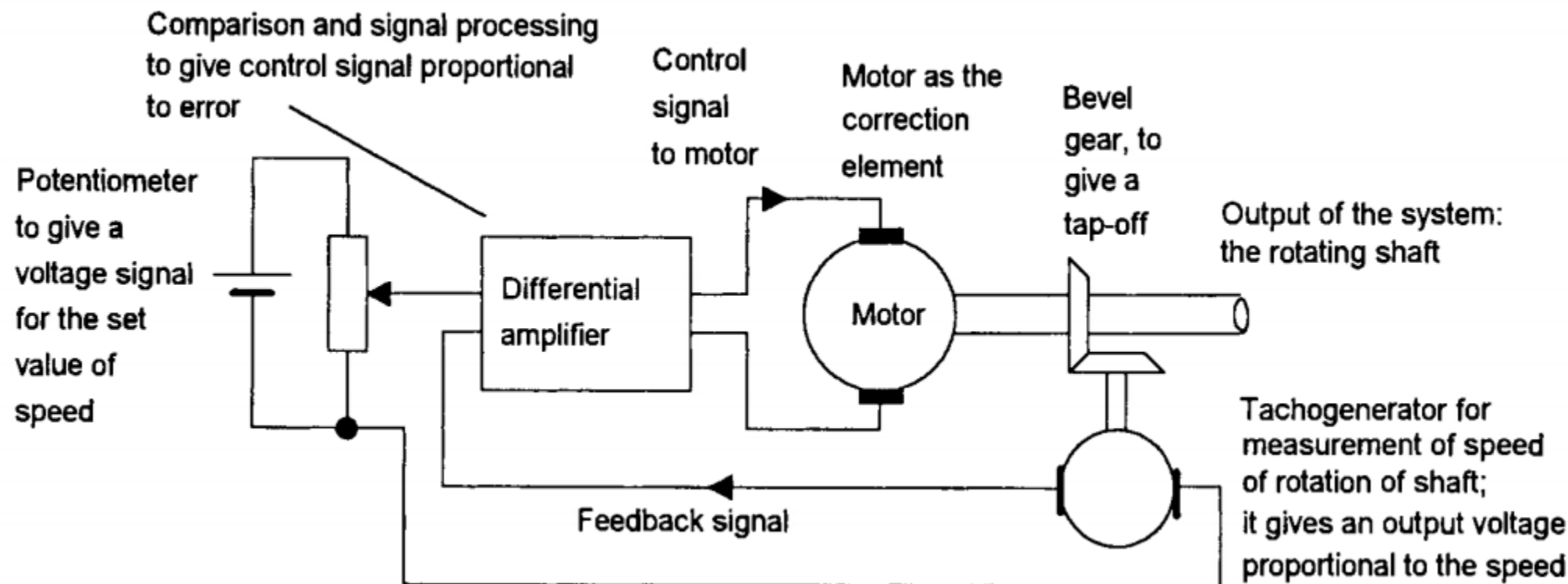


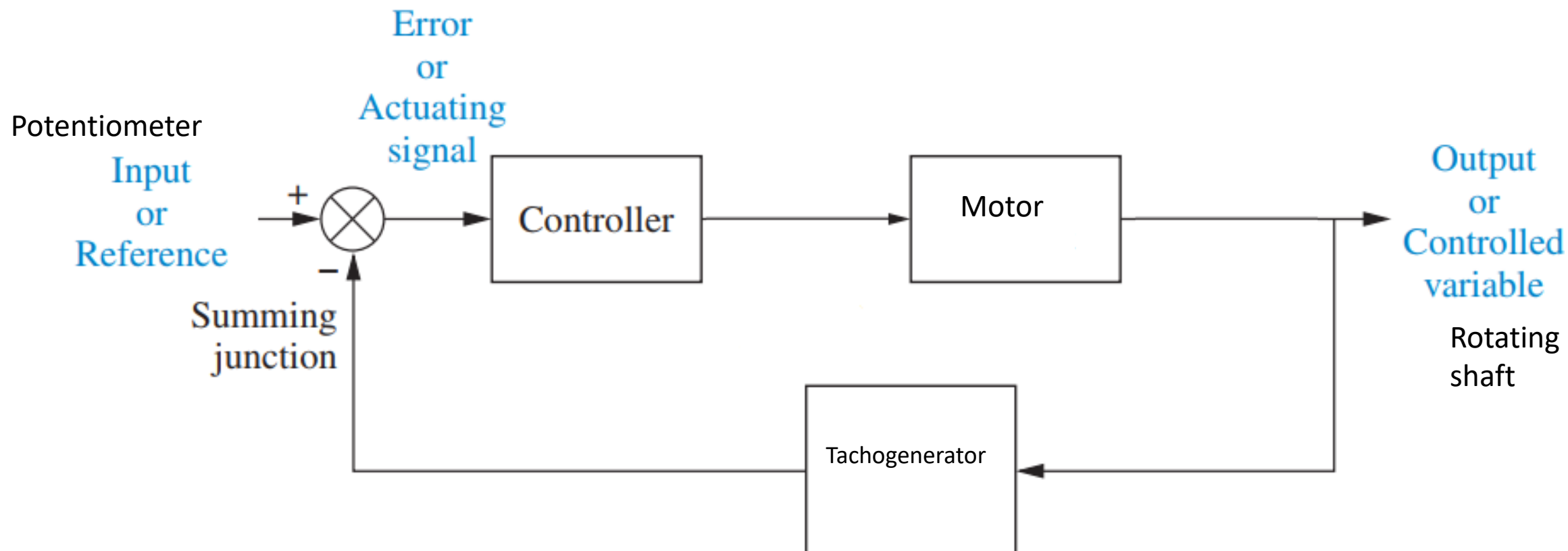
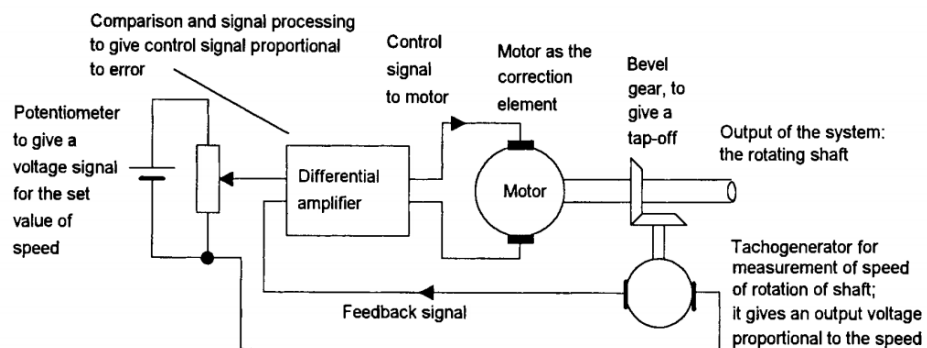
Closed-loop Control



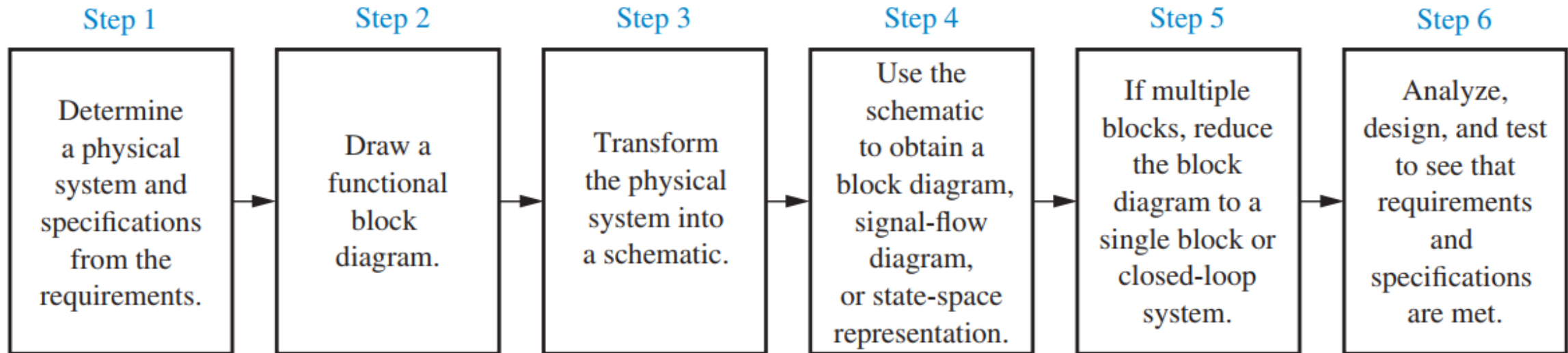
Example

Control of the speed of rotation of a motor shaft





The Design Process



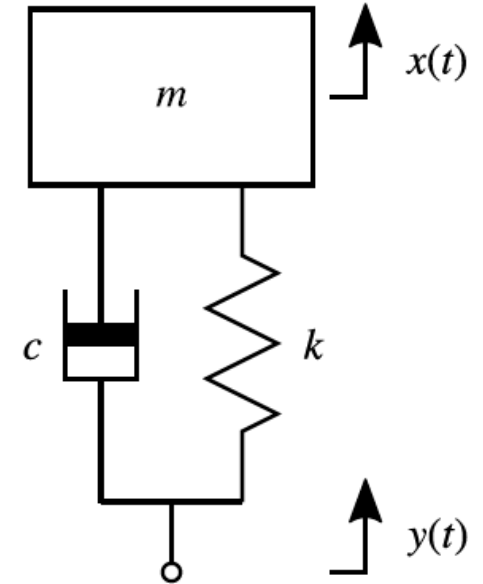
Introduction to Block Diagrams

- Use for **time/frequency**-domain **analysis** and **design**
- Graphical representation of the interconnections between the **components** of the system and the **flow** of signals
- Diagram is composed of **functional blocks**

Example:

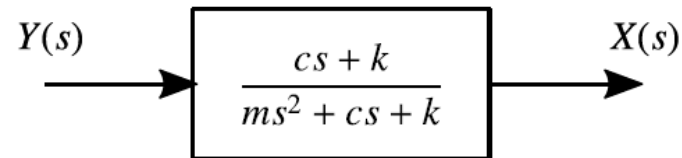
Block Diagram Representation of Control Systems

- Consider the familiar system shown, which has input $y(t)$ and response $x(t)$. We know that the equation of motion of this system is:
- $m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$
- Applying the Laplace transform with zero initial conditions:
- $(ms^2 + cs + k)X(s) = (cs + k)Y(s)$
- We then have the transfer function
- $\frac{X(s)}{Y(s)} = \frac{cs+k}{ms^2+cs+k}$



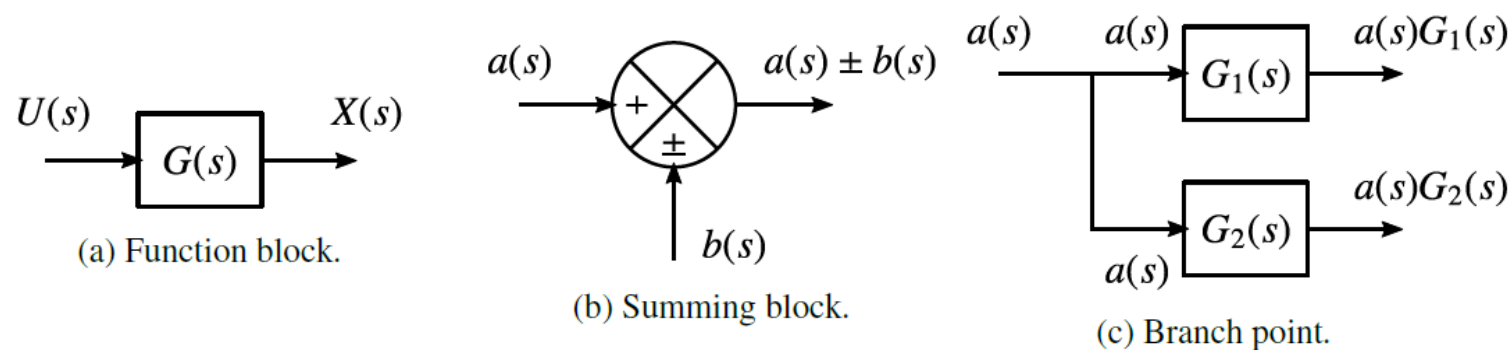
Transfer Function Blocks

- The transfer function
- $\frac{X(s)}{Y(s)} = \frac{cs+k}{ms^2+cs+k}$
- can be represented in block diagram form as shown in the figure below.
- This functional block is the basic element of a block diagram.



Block Diagram Elements

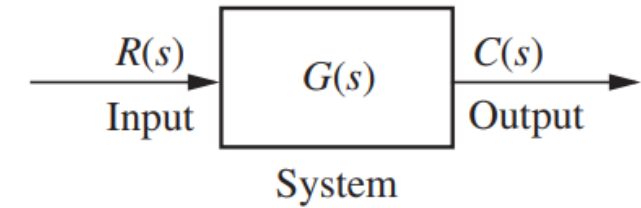
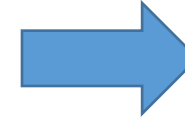
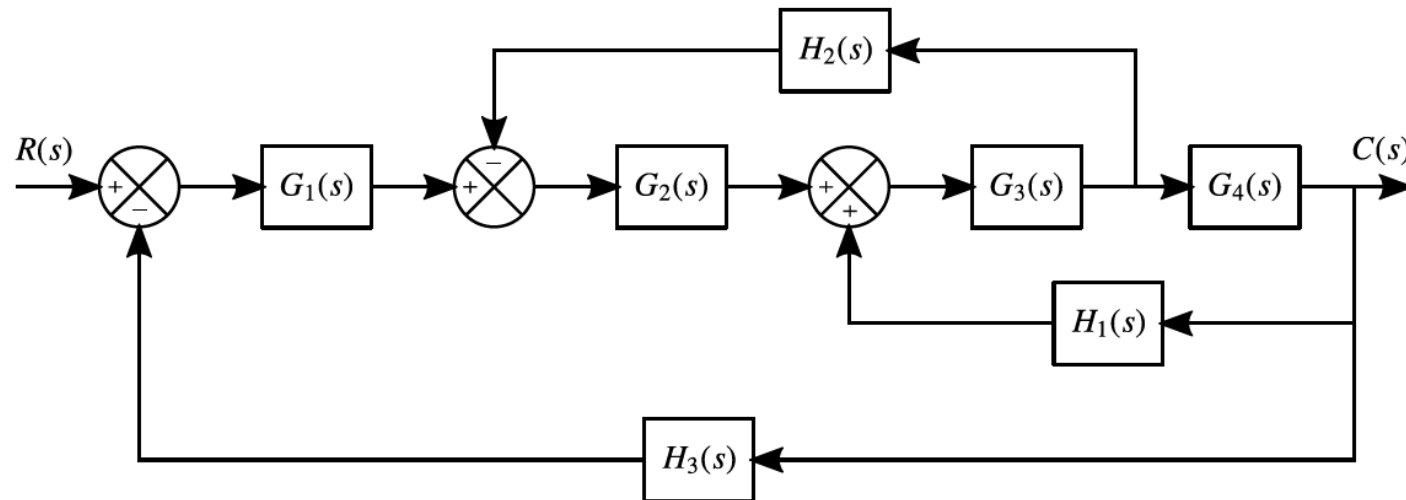
- In general terms, an element of a system with input $u(t)$, response $x(t)$ and transfer function $G(s)$ can be represented as shown in the bottom left figure.
- If the system is composed of several components, the blocks are connected by arrows showing the direction of **signal flow**.
- Where signals are compared (by addition or subtraction) is known as a **summing point**. A **branch point** is used to allow signals to go concurrently to other blocks or summing points.



Block Diagram Reduction

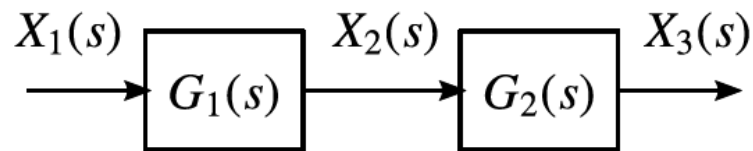
- Control systems can be complex in structure and design and naturally the resulting block diagram may also be complex.
- The block diagram will show all inputs, outputs and disturbances to the system.
- It is possible to reduce the block diagram to a simple transfer function relating a single input (or disturbance) to a single output.
- This process of reduction is carried by application of seven basic rules.

Block Diagram Reduction

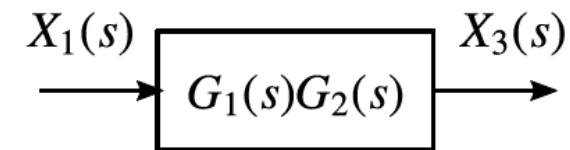


1: Cascaded Blocks

Original



Equivalent



Original:

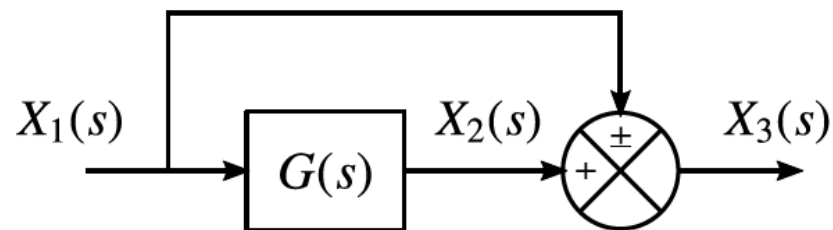
$$X_2(s) = G_1(s)X_1(s) \text{ and } X_3(s) = G_2(s)X_2(s), \\ \text{giving } X_3(s) = G_2(s)G_1(s)X_1(s)$$

Equivalent:

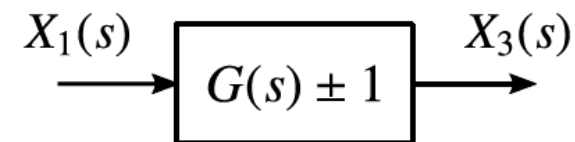
$$X_3(s) = G_2(s)G_1(s)X_1(s)$$

2: Summing Two Signals

Original



Equivalent



Original:

$$X_2(s) = X_1(s)G(s) \text{ and } X_3(s) = X_1(s) \pm X_2(s)$$

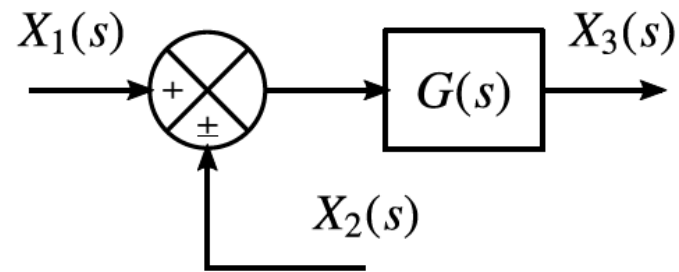
$$\text{giving } X_3(s) = X_1(s)(G(s) \pm 1)$$

Equivalent:

$$X_3(s) = X_1(s)(G(s) \pm 1)$$

3: Moving a Summing Point Behind a Block

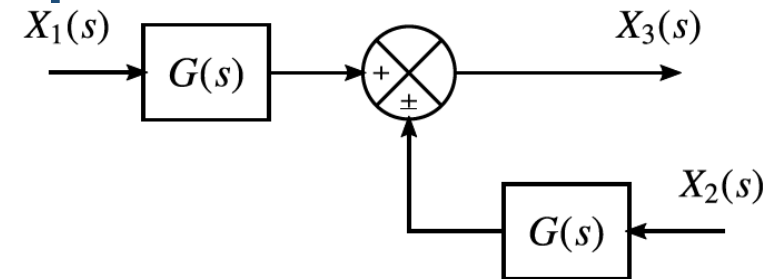
Original



Original:

$$X_3(s) = (X_1(s) \pm X_2(s))G(s)$$

Equivalent

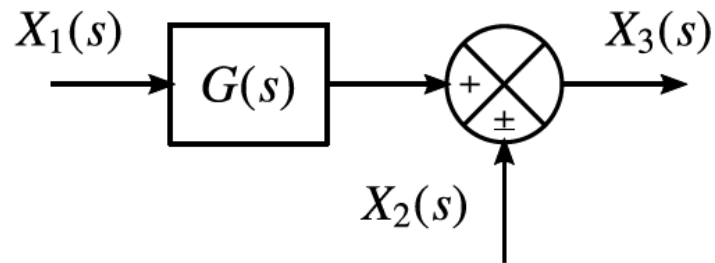


Equivalent:

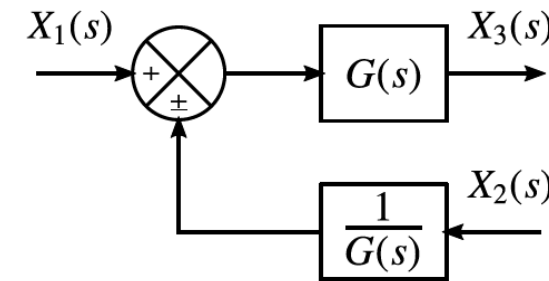
$$X_3(s) = X_1(s)G(s) \pm X_2(s)G(s)$$

4: Moving a Summing Point Ahead of a Block

Original



Equivalent



Original:

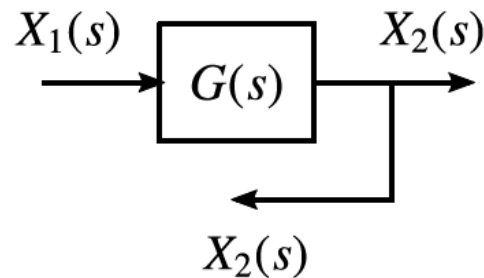
$$X_3(s) = X_1(s)G(s) \pm X_2(s)$$

Equivalent:

$$X_3(s) = \left(X_1(s) \pm \frac{X_2(s)}{G(s)} \right) G(s)$$

5: Moving a Branch Point Ahead of a Block

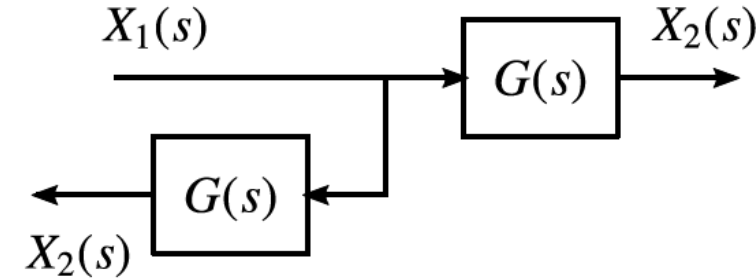
Original



Original:

$$X_2(s) = X_1(s)G(s)$$

Equivalent

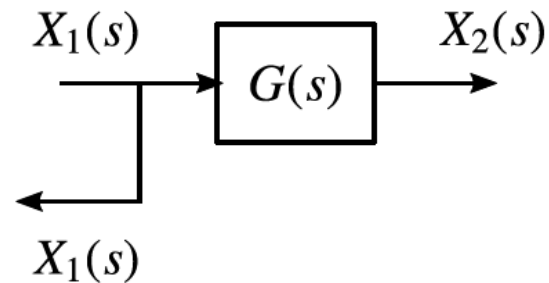


Equivalent:

$$X_2(s) = X_1(s)G(s)$$

6: Moving a Branch Point Behind a Block

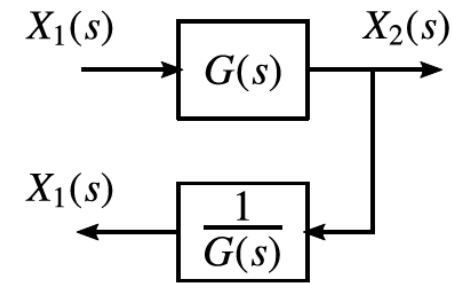
Original



Original:

$$X_2(s) = X_1(s)G(s) \text{ (RH signal), } X_1(s) \text{ (LH signal)}$$

Equivalent

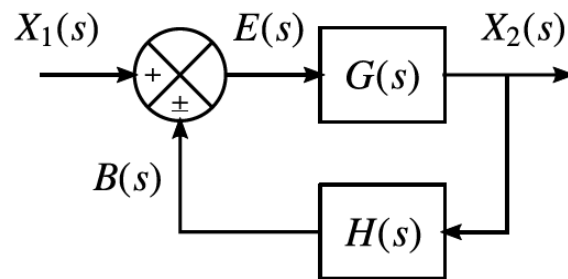


Equivalent:

$$X_2(s) = X_1(s)G(s) \text{ (RH signal), } \frac{X_2(s)}{G(s)} = X_1(s) \text{ (LH signal)}$$

7: Eliminating a Feedback Loop

Original



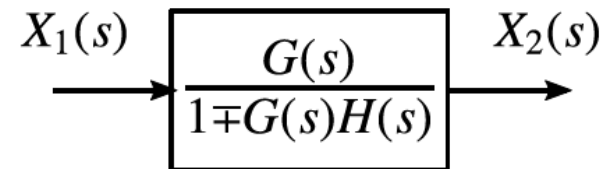
Original:

$$\begin{aligned}
 E(s) &= X_1(s) \pm B(s), \\
 X_2(s) &= E(s)G(s) \Rightarrow X_2(s) = (X_1(s) \pm B(s))G(s) \\
 B(s) &= H(s)X_2(s) \Rightarrow X_2(s) = (X_1(s) \pm X_2(s)H(s))G(s) \\
 &\Rightarrow (1 \mp G(s)H(s))X_2(s) = X_1(s)G(s)
 \end{aligned}$$

Equivalent:

$$\frac{X_2(s)}{X_1(s)} = \frac{G(s)}{1 \mp G(s)H(s)}$$

Equivalent



Summary

- There are two types of controllers:
 - Open-loop
 - Closed-loop
- A system, with or without control, can be represented by a **block diagram**
- A complex block diagram can be reduced to describe the system as a single **transfer function**

References:

-Control Systems Engineering, 7th Edition, N.S. Nise
-UESTC3001 2019/20 Notes, J. Le Kernec